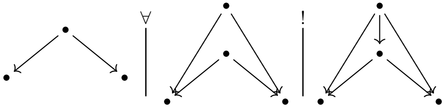
Categories, Allegories

Freyd & Scedrov, chapter 1 from §1.4 —

definitions, diagrams, theorems

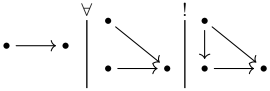
1.4 Cartesian categories

1.41 **MONIC PAIR** — ! reads "there is at most one extension to":



in symbols: $\square x = \square y \wedge \forall_{u,v}[(ux = vx) \wedge (uy = vy)] \implies u = v$.

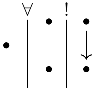
a single **MONIC** x, written $A \rightarrowtail B$ — variously called monomorphism, mono, injection; here always monic:



1.412 monic family: $\forall_{x \in \mathcal{F}} ux = vx \implies u = v$. A **TABLE** $\langle T; x_1, ..., x_n \rangle$ is the **TOP** T with a monic finite sequence out of it; the targets are the **FEET**, the x_i the **COLUMNS**.

an isomorphism class of tables is a **RELATION**, $\mathcal{R}e\ell(A_1, ..., A_n)$ the family of them on given feet. $n = 1$: a **SUBOBJECT** of A, family $\mathcal{S}u\ell(A)$. $n = 0$: a **VALUE**, family $\mathcal{V}a\ell$.

SUB-TERMINATOR — the top of a columnless table:

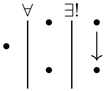


1.413 **CONTAINED**, $\langle T; x_i \rangle \subset \langle T'; x_i' \rangle$: some $z : T \to T'$ with $zx_i' = x_i$ — necessarily unique and monic. Containment pre-orders tables and partially orders relations, mutual containment being isomorphism; $\mathcal{R}e\ell$, $\mathcal{S}u\ell$, $\mathcal{V}a\ell$ are posets.

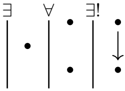
1.414 $A \to B$ is monic in $\boldsymbol{A}$ iff it is a sub-terminator of $\boldsymbol{A}/B$. Hence $\mathcal{S}u\ell_{\boldsymbol{A}}(B) \simeq \mathcal{V}a\ell_{\boldsymbol{A}/B}$.

1.415 **Rel** a table on A, B lists, without repetition, the instances of a relation from A to B — the extensional and the categorical notions coincide.

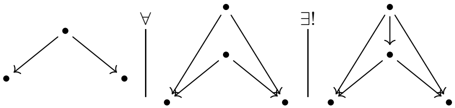
1.421 **TERMINATOR** (elsewhere: final, terminal), written 1, with $p_A : A \to 1$; a sub-terminator, representing the maximum value, unique up to a unique isomorphism:



a category has a terminator if

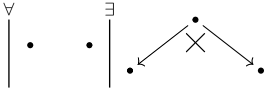


1.423 binary **PRODUCT** diagram:



written $A \times B$ with $\ell : A \times B \to A$, $\varkappa : A \times B \to B$; $\langle f, g\rangle$ the unique map with $\langle f, g\rangle\ell = f$, $\langle f, g\rangle\varkappa = g$; and $(x \times y) = \langle \ell' x, \varkappa' y\rangle$. $A \times B$ is a table on A, B and the maximum of $\mathcal{Rel}(A, B)$, so products are unique up to isomorphism.

a category has binary products if — the cross marking a product diagram:



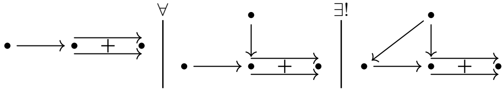
1.424 **Rel** products are the disjoint sums. In a poset, products are greatest lower bounds.

1.425 indexed product: P with $\{p_i : P \to A_i\}_I$ through which every $\{x_i : X \to A_i\}$ factors uniquely, $zp_i = x_i$; written $\Pi_I A_i$. Empty product = terminator; finite products = terminator + binary products. A poset has all finite products iff it is a semi-lattice.

1.426
$$\begin{tikzcd} & T \arrow[dl, "x"'] \arrow[dr, "y"] & \\ A & & B \end{tikzcd}$$

is a table iff $\langle x, y\rangle : T \to A \times B$ is monic, so $\mathcal{Rel}(A, B) \simeq \mathcal{Sub}(A \times B)$, generally $\mathcal{Rel}(A_1, \ldots, A_n) \simeq \mathcal{Sub}(A_1 \times \cdots \times A_n)$, and $\mathcal{Val} = \mathcal{Sub}(1)$.

1.428 **EQUALIZER** diagram — the puncture mark removes just the one equation, so the square carrying it need not commute:

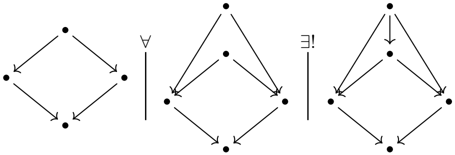


equalizers are monic, so they represent subobjects, and are essentially unique. In $\mathcal{S}$: $\{x \mid fx = gx\}$.

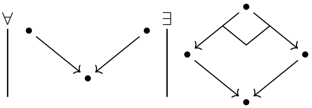
1.429 if $e, 1_A$ has an equalizer for every idempotent e, then *all idempotents split.*

1.43 **CARTESIAN CATEGORY**: finite products and equalizers (elsewhere: finitely complete, left-exact).

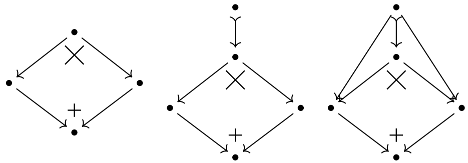
1.431 **PULLBACK** diagram — its upper half is a monic pair, and the relation that pair represents is fixed by the lower half:



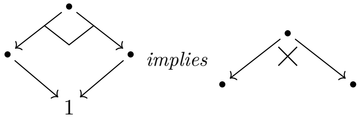
a category has pullbacks if — the corner mark ⌐ standing for the whole sentence above:



1.432 *Binary products + equalizers ⇒ pullbacks* — construct in sequence; the outer diagram on the right is the pullback:



1.433 *Pullbacks + a terminator ⇒ binary products*:



1.434 *Binary products + pullbacks ⇒ equalizers* — given $x, y : A \to B$, then $u = v$ and u equalizes x, y:

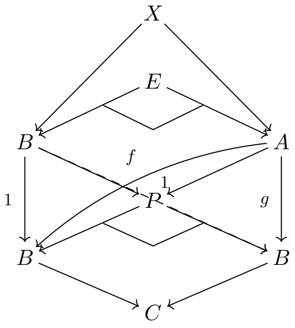
$$\begin{tikzcd}
& E \arrow[dl, "u"'] \arrow[dr, "v"] \arrow[dd, phantom, "\lrcorner" rotate=-45, very near start] & \\
A \arrow[dr, "\langle 1, x\rangle"'] & & A \arrow[dl, "\langle 1, y\rangle"] \\
& A \times B &
\end{tikzcd}$$

1.435–6 combining the last two: *pullbacks and a terminator imply cartesian.* The three lemmas are exhaustive — all sixteen combinations of terminator / binary products / equalizers / pullbacks not forced by them are realized.

1.437 a functor preserving finite products and equalizers is a **REPRESENTATION OF CARTESIAN CATEGORIES** such preserve pullbacks. Preserving pullbacks + the terminator suffices; so does preserving finite products + pullbacks of monics.

1.438 *A functor reflecting equalizers reflects isomorphisms* ($A \to B$ is iso iff it equalizes $1_B, 1_B$). From a category with equalizers, an isomorphism-reflecting equalizer-preserving functor is an embedding, hence faithful. A faithful functor preserving terminators / products / equalizers / pullbacks reflects them too.

1.439 *With pullbacks, if* $A \overset{f,g}{\to} B \to C$ *then* f, g *have an equalizer* — $P = B \times_C B$ replaces $B \times B$, and $E \to A$ is the equalizer:



A functor from a cartesian category preserving pullbacks preserves equalizers.

1.44 $\Sigma : \boldsymbol{A}/B \to \boldsymbol{A}$ does not preserve terminators — $\boldsymbol{A}/B$ has the distinguished terminator 1_B, carried to B — and is universal in that: for $T : \boldsymbol{C} \to \boldsymbol{A}$ with $T(1) = B$ there is a unique $T' : \boldsymbol{C} \to \boldsymbol{A}/B$ with $T'(1) = 1_B$ and $T = T'\Sigma$.

specializing to $T = B \times -$ gives the **DIAGONAL FUNCTOR** $\Delta(A) = \left(B \times A \overset{\varkappa}{\to} B\right)$ and the factorization $\boldsymbol{A} \overset{\Delta}{\to} \boldsymbol{A}/B \overset{\Sigma}{\to} \boldsymbol{A}$.

1.441 if $\boldsymbol{A}$ has pullbacks then $\boldsymbol{A}/B$ is cartesian and Σ preserves pullbacks and equalizers; Σ is always faithful.

1.442 the Cayley representation preserves and reflects pullbacks and equalizers; factoring it as $\boldsymbol{A} \overset{C'}{\to} \mathcal{S}^{|\boldsymbol{A}|} \overset{\Sigma}{\to} \mathcal{S}$ gives a faithful representation. *Every small cartesian category is faithfully represented in a power of $\mathcal{S}$.*

the A-th coordinate of C' is the **REPRESENTABLE FUNCTOR** $(A, -)$: $B \mapsto (A, B)$, $f \mapsto (A, f)$, i.e. $g \mapsto gf$. *Representable functors from a cartesian category are representations of cartesian categories, and are collectively faithful.*

1.443 a universal sentence in the pullback / equalizer predicates with a counterexample anywhere has one in $\mathcal{S}$: cut down to a small subcategory, then apply Cayley.

1.444 **HORN SENTENCE**: a universally quantified $(P_1 \wedge \dots \wedge P_n) \Longrightarrow Q$ with all P_i, Q primitive; for cartesian categories the primitives are the categorical ones plus "is a terminator / product / equalizer". *Any Horn sentence in the theory of cartesian categories true for $\mathcal{S}$ is true for every cartesian category* — apply a collectively faithful $(A, -)$ to the counterexample.

1.45 pullbacks transfer monics:

$$\begin{tikzcd} \bullet \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \bullet \arrow[d] \\ \bullet \arrow[r, tail] & \bullet \end{tikzcd} \quad \textit{implies} \quad \begin{tikzcd} \bullet \arrow[r, tail] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \bullet \arrow[d] \\ \bullet \arrow[r, tail] & \bullet \end{tikzcd}$$

that is, if

$$\begin{tikzcd} {} \arrow[r, "\overline{x}"] \arrow[d] & {} \arrow[d] \\ {} \arrow[r, "x"'] & {} \end{tikzcd}$$

is a pullback and x is monic, then so is $\overline{x}$.

1.451 $A_1 \rightarrowtail A$ is an **INVERSE IMAGE** of $B_1 \rightarrowtail B$ along f:

$$\begin{tikzcd} A_1 \arrow[r, tail] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & A \arrow[d, "f"'] \\ B_1 \arrow[r, tail] & B \end{tikzcd}$$

inverse images preserve containment, hence are well defined on subobjects: an order-preserving $f^{\#} : \mathcal{Sub}(B) \to \mathcal{Sub}(A)$, making $\mathcal{Sub}(-)$ a contravariant poset-valued functor.

1.452 pulling back two monics gives their greatest lower bound in $\mathcal{Sub}(A)$:

$$\begin{tikzcd} & A_{12} \arrow[dl, tail] \arrow[dr, tail] \arrow[dd, phantom, "\lrcorner" rotate=-45, very near start] & \\ A_1 \arrow[dr, tail] & & A_2 \arrow[dl, tail] \\ & A & \end{tikzcd}$$

so $\mathcal{Sub}(A)$ is a **SEMI-LATTICE** — a poset with finite intersections, the empty one 1_A — and $\mathcal{Sub}(-)$ lands in semi-lattices, inverse images preserving intersections.

1.453 **LEMMA** *$\boldsymbol{A}$ cartesian, T preserving pullbacks: T is faithful iff T preserves properness of subobjects* — if $A' \rightarrowtail A$ is not iso, neither is $TA' \rightarrowtail TA$.

1.454 the kernel-pair of f measures how far f is from monic: the **LEVEL** of f (elsewhere: kernel-pair, congruence). f is monic iff Δ is iso; this monic $\Delta : A \to L$ is the **DIAGONAL** morphism, representing the diagonal subobject.

$$\begin{tikzcd}
& A \arrow[d, tail, "\Delta"'] \arrow[ddl, bend right, "1"'] \arrow[ddr, bend left, "1"] & \\
& L \arrow[dl, "l"'] \arrow[dr, "r"] \arrow[dd, phantom, "\ulcorner", very near start] & \\
A \arrow[dr, "f"'] & & A \arrow[dl, "f"'] \\
& B &
\end{tikzcd}$$

1.46 the cartesian structure of the prime examples.

1.462 for small $\boldsymbol{A}$, $\mathcal{S}^{\boldsymbol{A}}$ is cartesian and the **EVALUATION FUNCTORS** $E_A : \mathcal{S}^{\boldsymbol{A}} \to \mathcal{S}$, $E_A(F) = F(A)$, are representations of cartesian categories, collectively faithful. Assembled into the forgetful $\mathcal{S}^{\boldsymbol{A}} \to \mathcal{S}^{|\boldsymbol{A}|}$ they give a faithful representation into a power of $\mathcal{S}$. Hence $F_1 \to F_2$ is monic iff every $F_1 A \to F_2 A$ is.

1.463 H^A denotes $(A, -)$ viewed in $\mathcal{S}^{\boldsymbol{A}}$. As a right $\boldsymbol{A}$-set it is the sub $\boldsymbol{A}$-set of $\boldsymbol{A}$ generated by 1_A: for any right $\boldsymbol{A}$-set X and $x \in X$ with $x\Box = A$ there is a unique $H^A \to X$ sending $1_A \mapsto x$, $y \mapsto xy$. So $\left(H^A, X\right) \simeq \{x \in X \mid x\Box = A\}$, and for the functor F associated to X, $\left(H^A, F\right) \simeq F(A)$, naturally — $\left(H^A, -\right)$ and E_A are conjugate.

1.464 $\left(H^A, H^B\right) \simeq (B, A)$, so each $x : B \to A$ gives $H^x : H^A \to H^B$ and a contravariant full embedding $H : \boldsymbol{A} \to \mathcal{S}^{\boldsymbol{A}}$, the **YONEDA REPRESENTATION**.

Hom : $\boldsymbol{A}^\circ \times \boldsymbol{A} \to \mathcal{S}$ sends A, B to (A, B); the composite $\boldsymbol{A} \times \mathcal{S}^{\boldsymbol{A}} \xrightarrow{H^\circ \times 1} \left(\mathcal{S}^{\boldsymbol{A}}\right)^\circ \times \mathcal{S}^{\boldsymbol{A}} \xrightarrow{\text{Hom}} \mathcal{S}$ is naturally equivalent to $E : A, F \mapsto F(A)$.

1.465 H_B denotes the contravariant $(-, B)$ viewed in $\mathcal{S}^{\boldsymbol{A}^\circ}$; again $(H_B, F) \simeq F(B)$, giving a covariant full embedding $\boldsymbol{A} \to \mathcal{S}^{\boldsymbol{A}^\circ}$, also called the yoneda representation. It preserves and reflects the cartesian predicates.

1.47 $\boldsymbol{A}$ is *special* if every universal (not merely Horn) sentence in the cartesian predicates true for $\mathcal{S}$ holds in $\boldsymbol{A}$.

1.471–4 *A special cartesian category has at most two values.* If every finite sequence of morphisms is preserved-and-reflected by some $T : \boldsymbol{A} \to \mathcal{S}$, then $\boldsymbol{A}$ is special. One-valued $\boldsymbol{A}$: special iff every $B \times -$ is faithful. Two-valued $\boldsymbol{A}$, with 0 the proper subobject of 1: special iff $B \times -$ is faithful for every $B \not\cong 0$.

1.5 Regular categories

1.51 a subobject $B' \rightarrowtail B$ **ALLOWS** f iff f factors through it:

$$\begin{tikzcd} & A \arrow[dl] \arrow[d, "f"'] \\ B' \arrow[r, tail] & B \end{tikzcd}$$

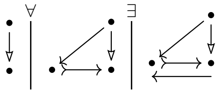
with pullbacks: B' allows f iff $f^{\#}(B')$ is entire, all of A. A faithful pullback-preserving functor reflects allowance.

the **IMAGE** of f is the smallest subobject allowing f; a category **HAS IMAGES** if every morphism has one. Then $f : \mathcal{Sub}(A) \to \mathcal{Sub}(B)$ sends $A' \rightarrowtail A$ to the image of $A' \rightarrowtail A \xrightarrow{f} B$.

with pullbacks too, the first adjoint pair of poset functions: $f(A') \subset B'$ iff $A' \subset f^{\#}(B')$. f is the **LEFT-ADJOINT** of $f^{\#}$, $f^{\#}$ the **RIGHT-ADJOINT** of f.

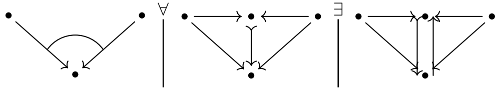
1.511 *If $\boldsymbol{A}$ has images and T is faithful and preserves images, then T reflects images.*

1.512 $A \to B$ is a **COVER**, written $A \twoheadrightarrow B$, if its image is entire — every monic it factors through is invertible:



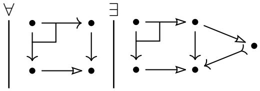
covers are closed under composition and left cancellation ($A \to B \to C$ a cover forces $B \to C$ to be one); a monic cover is an isomorphism.

1.513 $\{A_i \to B\}$ **COVERS** if no proper subobject of B allows all of them — the arc across the two arrows says the pair covers:



1.514 $\{A_i \to B\}$ is **EPIC** if its members collectively cancel — a monic family in the opposite category. With equalizers, cover implies epic; the converse is a special property.

1.52 a **REGULAR CATEGORY** is a cartesian category with images in which pullbacks transfer covers — one sentence for both clauses, the right-hand vertex being the image of the right leg:

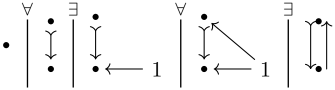


a **PRE-REGULAR CATEGORY** is cartesian, with or without images, and transfers covers by pullback — the strength at which the statements below hold. A **REPRESENTATION OF PRE-REGULAR CATEGORIES** preserves finite products, equalizers and covers, hence whatever images exist.

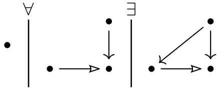
1.521 for small $\boldsymbol{A}$, $\mathcal{S}^{\boldsymbol{A}}$ is regular and the evaluation functors are collectively faithful representations of regular categories.

1.522 the **SUPPORT** $\mathcal{S}pt(A)$ is the image of $A \to 1$; A is **WELL-SUPPORTED** if $\mathcal{S}pt(A) = 1$ — pre-regularly, if $A \to 1$ covers.

1.523 A is **WELL-POINTED** if all of $1 \to A$ covers A: no proper subobject of A allows every point.



1.524 P is **PROJECTIVE** if $(P, -)$ preserves covers — every $P \to B$ lifts through every cover:



Pre-regularly it suffices that every cover targeted at P have a left inverse.

1.525 a pre-regular category is **CAPITAL** if every well-supported object is well-pointed. In a capital pre-regular category the terminator is projective.

1.526 so for capital pre-regular $\boldsymbol{A}$ the functor Γ represented by 1 is a representation of pre-regular categories.

1.53 **THE SLICE LEMMA** factor $(B \times -)$ as $\boldsymbol{A} \xrightarrow{\Delta} \boldsymbol{A}/B \xrightarrow{\Sigma} \boldsymbol{A}$, Δ the diagonal functor [1.44], $\Delta(1) = \left(B \xrightarrow{1} B\right)$. *If $\boldsymbol{A}$ is (pre-)regular so is $\boldsymbol{A}/B$; Δ is a representation of pre-regular categories; Δ is faithful iff B is well-supported.*

1.531 $\Sigma : \boldsymbol{A}/B \to \boldsymbol{A}$ preserves and reflects covers and pullbacks and carries factorizations back, so images in $\boldsymbol{A}$ give images in $\boldsymbol{A}/B$.

1.535 *Diagonal* also names the morphism $A \to \Pi_I A$ with identity coordinates; Δ is not that functor but is conjugate to it.

1.54 **THE CAPITALIZATION LEMMA** let $\mathcal{C}$ have small pre-regular categories as objects and faithful representations as morphisms.

1.541 $\mathcal{C}$ meets the *equivalence condition* if $\boldsymbol{A} \in \mathcal{C}$ and an equivalence $\boldsymbol{A} \to \boldsymbol{B}$ force $\boldsymbol{B}, (\boldsymbol{A} \to \boldsymbol{B}) \in \mathcal{C}$; the *slice condition* if $\boldsymbol{A} \in \mathcal{C}$ and B well-supported force $\boldsymbol{A}/B, \Delta \in \mathcal{C}$; the *union condition* if a **DIRECTED UNION** $\boldsymbol{A} = \cup \mathcal{F}$ of $\mathcal{C}$-subcategories with $\mathcal{C}$-inclusions has $\boldsymbol{A}, (\boldsymbol{A}' \subset \boldsymbol{A}) \in \mathcal{C}$.

1.542 all three hold for the small pre-regular and the small regular categories, the slice condition being [1.53].

1.543 *Under the three conditions every $\boldsymbol{A} \in \mathcal{C}$ admits a capital $\underline{\underline{\boldsymbol{A}}} \in \mathcal{C}$ with $\boldsymbol{A} \to \underline{\underline{\boldsymbol{A}}}$ faithful in $\mathcal{C}$.*

1.544 to read $\boldsymbol{A}$ inside $\boldsymbol{A}/B$, pass to the **INFLATION** $\boldsymbol{A}'$ — objects finite sequences of objects, binary product concatenation — giving strict cancellation $B \times A = B \times A' \implies A = A'$, then replace the image of Δ by $\boldsymbol{A}$.

1.545 $\boldsymbol{A} \subset \boldsymbol{A}^*$ is a **RELATIVE CAPITALIZATION** if every proper $B' \rightarrowtail B$ of $\boldsymbol{A}$, B well-supported, fails to allow some $x : 1 \to B$ of $\boldsymbol{A}^*$. Iterated union gives $\underline{\underline{\boldsymbol{A}}}$; each slicing step supplies ΔB with its generic point, which $\Delta B'$ cannot allow:

$$\begin{tikzcd} B \arrow[r, "\langle 1,1\rangle"] \arrow[dr, "1"'] & B \times B \arrow[d] \\ & B \end{tikzcd}$$

1.547 choice-free variant: the rational category [1.48] of pairs $\langle A, F\rangle$, F a finite set of morphisms to distinct well-supported targets — a directed union of slices $\boldsymbol{A}/\Pi U$.

1.55 **THE HENKIN–LUBKIN THEOREM** *Every small pre-regular category is faithfully represented in a power of the category of sets:* $T_B = \boldsymbol{A} \xrightarrow{\Delta} \boldsymbol{A}/B \to \underline{\underline{\boldsymbol{A}/B}} \xrightarrow{\Gamma} \mathcal{S}$; for regular $\boldsymbol{A}$ the $\{T_U\}_{U \subset 1}$ already suffice.

1.551 *Every Horn sentence in the defining predicates of regular categories true for the category of sets is true for every regular category.*

1.552 a pre-regular category is **SPECIAL** if every universally quantified sentence in the relevant predicates true for $\mathcal{S}$ holds in it; the conditions of [1.47] stay necessary. *Special iff* $A \to U \rightarrowtail 1$ *forces* $A \to U$ *or* $U \rightarrowtail 1$ *to be an isomorphism*

equivalently: one-valued ($|\pi_0 \boldsymbol{A}| = 1$), or two-valued ($|\pi_0 \boldsymbol{A}| = 2$) with every object well-supported or isomorphic to 0, the unique proper sub-terminator.

1.56 tables T_1 on A, B and T_2 on B, C: a table T_{12} on A, C is a **COMPOSITION** if there exists

$$\begin{tikzcd}
 & & T_{12} \arrow[dddll] \arrow[dddrr] & & \\
 & & \cdot \arrow[u] \arrow[dl] \arrow[dr] & & \\
 & T_1 \arrow[dl] \arrow[dr] & & T_2 \arrow[dl] \arrow[dr] & \\
A & & B & & C
\end{tikzcd}$$

cartesian + images builds one exactly so; shrinking either factor shrinks the composite, giving an order-preserving $\mathcal{R}\mathit{el}(A,B) \times \mathcal{R}\mathit{el}(B,C) \to \mathcal{R}\mathit{el}(A,C)$. *Caution:* not necessarily associative [1.569].

1.561 the **RECIPROCAL** R° of R from A to B twists the columns: $R^{\circ\circ} = R$, $(RS)^\circ = S^\circ R^\circ$. Reciprocation reverses composition, preserves the ordering.

1.562 $\mathcal{R}\mathit{el}(A,B) \simeq \mathcal{S}\mathit{ub}(A \times B)$ makes $\mathcal{R}\mathit{el}(A,B)$ a semi-lattice: $(R \cap S)T \subset RT \cap ST$, $(R \cap S)^\circ = S^\circ \cap R^\circ$.

1.563 *F preserving the cartesian structure and images induces $\mathscr{Rel}(A,B) \to \mathscr{Rel}(FA,FB)$ preserving composition, reciprocation and intersection; faithful F reflects them.*

Hence a Horn sentence on maps and relations — the regular predicates, plus composition, reciprocation and intersection — true of binary relations between sets is true in every regular category.

in particular $R(ST) = (RS)T$, and the **MODULAR IDENTITY** $RS \cap T \subset (R \cap TS^\circ)S$. Every containment is an equation: $R \subset S$ iff $R = R \cap S$.

1.564 $\mathscr{Rel}(\boldsymbol{A})$ is the category of relations; the **GRAPH** of x is tabulated by

$$\begin{tikzcd} & A \arrow[dl, "1"'] \arrow[dr, "x"] & \\ A & & B \end{tikzcd}$$

giving $\boldsymbol{A} \to \mathscr{Rel}(\boldsymbol{A})$. Treat $\boldsymbol{A}$ as a subcategory: **MAP** means a morphism of $\mathscr{Rel}(\boldsymbol{A})$ lying in $\boldsymbol{A}$, and $\langle T; x, y\rangle$ tabulates one iff x is an isomorphism.

R is **ENTIRE** iff $1 \subset RR^\circ$, **SIMPLE** iff $R^\circ R \subset 1$. For R tabulated by x, y: x covers iff R is entire, x is monic iff R is simple. So R is a map iff entire and simple — hence as soon as every representation $F : \boldsymbol{A} \to \mathcal{S}$ makes $F(R)$ one.

1.565 a **PUSHOUT** is a pullback in the opposite category; the corner mark moves to the far vertex of the square, C.

In a regular category a pullback of covers is a pushout — all four edges then cover, the mediating map being the relation $(x^\circ u) \cap (y^\circ v)$, a map because it is one in $\mathcal{S}$:

1.566 a **COEQUALIZER** — an equalizer in the opposite category — is always a cover. *In a regular category every cover is a coequalizer* — of its own level:

$$\begin{tikzcd} & \bullet \arrow[dl, "l"'] \arrow[dr, "r"] \arrow[dd, phantom, "\lrcorner" very near start, rotate=-45] & \\ A \arrow[dr, "x"'] & & A \arrow[dl, "x"] \\ & B & \end{tikzcd}$$

1.567 an endo-relation E is an **EQUIVALENCE RELATION** if $1 \subset E$, $E^\circ \subset E$, $EE \subset E$. Levels are equivalence relations:

$$\begin{tikzcd} & \bullet \arrow[dl, "l"'] \arrow[dr, "r"] \arrow[dd, phantom, "\lrcorner" very near start, rotate=-45] & \\ A \arrow[dr, "x"'] & & A \arrow[dl, "x"] \\ & B & \end{tikzcd}$$

E is **EFFECTIVE** if it is the level of some map; an **EFFECTIVE REGULAR CATEGORY** is one where every equivalence relation is effective. The metatheorem [1.551] extends to these, $\mathcal{S}$ being effective.

1.568 covers out of A are preordered by $f \leq g$ iff f factors through g:

$$\begin{tikzcd} A \arrow[r, "g"] \arrow[dr, "f"'] & B \arrow[d, dashed] \\ & C \end{tikzcd}$$

the associated poset is $\mathcal{Quot}(A)$, of **QUOTIENT-OBJECTS** — not the dual of $\mathcal{Sub}(A)$. By [1.566] a faithful order-reversing functor sends it to the poset of equivalence relations on A.

1.569 *For* $\boldsymbol{A}$ *cartesian with images, composition of relations is associative iff* $\boldsymbol{A}$ *is regular* — x covers iff $x^\circ x = 1$; if z fails to cover, $(yx^\circ)x$ is not even a map.

1.56(10) x is a **CONSTANT MORPHISM** if $yx = y'x$ for all $y, y' : C \to A$. In a regular category its image is a sub-terminator.

1.56(11) *In a regular category an object is projective iff every entire relation out of it contains a map.*

1.57 an object is **CHOICE** if every entire relation targeted at it contains a map. Every object is projective iff every object is choice; an **AC REGULAR CATEGORY** is a regular category with either — equivalently, a cartesian category where every morphism is a left-invertible followed by a monic.

1.571 if $\boldsymbol{A}$ is cartesian and every $x : A \to B$ admits $e : A \to A$ with $e^2 = e$, $ex = x$, then e and x share a level and $\boldsymbol{A}$ is AC regular — the image of x appears as a subobject of A.

1.572 the category $\boldsymbol{R}$ of recursive functions on $0, 1, 2, \ldots, \omega$ is AC regular but not effective: a left inverse would choose representatives, forcing a recursively enumerable equivalence relation to be recursive.

1.58 a **BICARTESIAN CATEGORY** is both cartesian and **COCARTESIAN**, the latter meaning the opposite category is cartesian. Dual to a terminator is a **COTERMINATOR**, written 0; dual to a product a **COPRODUCT**, written $A + B$.

in a cartesian category an object 0 whose incoming morphisms are all isomorphisms is a coterminator, then called a **STRICT COTERMINATOR** — agreeing with the earlier 0 [1.474, 1.552].

1.581 between regular cocartesian categories a representation of bicartesian categories is a representation of regular categories: it preserves coequalizers, hence covers [1.566].

1.582 for bicartesian $\boldsymbol{A}$ regularity is a pair of Horn sentences in the bicartesian predicates — the image of x is $A \twoheadrightarrow C \rightarrowtail B$, with $A \twoheadrightarrow C$ the coequalizer of the level of x.

1.583 effectiveness likewise: for l, r tabulating an equivalence relation and x their coequalizer, E is effective iff

$$\begin{tikzcd}
 & E \arrow[dl, "l"'] \arrow[dr, "r"] & \\
A \arrow[dr, "x"'] & & A \arrow[dl, "x"] \\
 & C &
\end{tikzcd}$$

is a pullback.

1.584 $\boldsymbol{A}/B$ is cocartesian and Σ a faithful representation of cocartesian categories.

1.586 for small $\boldsymbol{A}$, $\mathcal{S}^{\boldsymbol{A}}$ is cocartesian with collectively faithful evaluation functors, hence effective regular.

1.587 no elementary characterization of the bicartesian categories obeying every Horn sentence true for $\mathcal{S}$: $1 + \mathbb{N} \xrightarrow{\binom{0}{s}} \mathbb{N}$ iso plus $\mathbb{N} \to 1$ a coequalizer of $1, s$ pins down $\mathbb{N}, 0, s$, then addition and multiplication; the equalizer of two polynomials is a coterminator iff the equation is unsolvable, so those Horn sentences encode the diophantine problems and are not recursively enumerable.

1.6 Pre-logoi

1.6 A **PRE-LOGOS**: a regular category in which $\mathcal{Sub}(A)$ is a lattice, not just a semi-lattice, for each A, and $f^{\#} : \mathcal{Sub}(B) \to \mathcal{Sub}(A)$ is a lattice homomorphism for each $f : A \to B$.

Elementary form: each pair of subobjects has a union, preserved under inverse image.

Equivalently: a cartesian category with images in which pullbacks transfer finite covers. For a cover $\{B_i \to B\}_{i=1}^{n}$ and a map $A \to B$ there exist

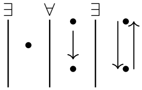
$$\begin{tikzcd} A_i \arrow[r] \arrow[d] & A \arrow[d] \\ B_i \arrow[r] & B \end{tikzcd}$$

$i = 1, \ldots, n$, with $\{A_i \to A\}_{i=1}^{n}$ a cover. Their existence forces the pullback squares to be such.

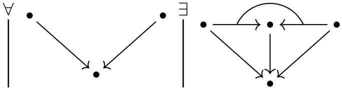
1.61 $0 =$ the minimal subobject of 1; $p_A^{\#}(0) =$ the minimal subobject of A. *Any morphism to* 0 *is an isomorphism.* Each A receives exactly one $0 \to A$: 0 is a coterminator.

A pre-logos is degenerate iff it is one-valued, $0 \simeq 1$.

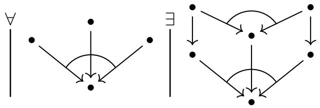
1.611 Hence a pre-logos is a cartesian category with images and three sentences. First, 0 exists and every morphism to it is invertible:



second, any two morphisms with a common target have a union — the arc says the pair covers it:



third, inverse image preserves unions — the squares of 1.6 again, now with both pairs covering:



1.612 For monic $f : A \rightarrowtail B$ the inverse image sends $B' \subset B$ to $A \cap B'$.

Every such $f^{\#}$ preserves binary unions iff $\mathcal{Sub}(B)$ is a **DISTRIBUTIVE LATTICE**: $A \cap (B_1 \cup B_2) = (A \cap B_1) \cup (A \cap B_2)$.

1.613 A poset viewed as a category is cartesian iff it is a semi-lattice (intersections are products, equalizers hold by default); a semi-lattice is regular, the only covers being identities. A semi-lattice is a pre-logos iff it is a distributive lattice.

1.614 A **REPRESENTATION OF PRE-LOGOI**: a functor between pre-logoi preserving the cartesian structure, images, and finite unions, the empty union included.

1.615 In a bicartesian category with images the union of $x_1 : A_1 \rightarrowtail A$, $x_2 : A_2 \to A$ is the image of $\binom{x_1}{x_2} : A_1 + A_2 \to A$.

Building on [1.582], a finite set of Horn sentences in the bicartesian predicates holds for a bicartesian category iff it is a pre-logos.

1.616 $\mathcal{Rel}(A, B) \simeq \mathcal{Sub}(A \times B)$ is a distributive lattice, so relations carry $R \cup S$.

$\mathcal{Sub}(A \times B)$ distributive	$R \cap (S \cup T) = (R \cap S) \cup (R \cap T)$
reciprocation	$(R \cup S)^\circ = S^\circ \cup R^\circ$
$yS = (y \times 1)^{\#}(S)$, inverse images	$y(S \cup T) = yS \cup yT$
$x^\circ(yS) = (x \times 1)(yS)$, direct images	$R(S \cup T) = RS \cup RT$
reciprocate the line above	$(S \cup T)R' = SR' \cup TR'$

1.62 **PASTING LEMMA** *For subobjects A_1, A_2 of A in a pre-logos,*

$$\begin{tikzcd} A_1 \cap A_2 \arrow[r] \arrow[d] & A_2 \arrow[d] \\ A_1 \arrow[r] & A_1 \cup A_2 \end{tikzcd}$$

is a pushout — the mediating map out of $A_1 \cup A_2$ is unique because A_1, A_2 cover it.

Restated: a pullback whose two morphisms into the lower right corner cover is a pushout.

$$\begin{tikzcd} \bullet \arrow[r, tail] \arrow[d, tail] \arrow[dr, phantom, "\lrcorner", very near start] & \bullet \arrow[d, tail] \\ \bullet \arrow[r, tail] & \bullet \end{tikzcd} \quad \textit{implies} \quad \begin{tikzcd} \bullet \arrow[r] \arrow[d] & \bullet \arrow[d] \\ \bullet \arrow[r] \arrow[ur, phantom, "\ulcorner", very near end] & \bullet \end{tikzcd}$$

1.621 *If* $A_1 \cap A_2 = 0$ *and* $A_1 \cup A_2 = A$ *then* A *is a coproduct of* A_1, A_2 — the special case of the square above.

1.623 A **POSITIVE PRE-LOGOS**: a pre-logos in which every pair A, B admits

$$\begin{tikzcd} 0 \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & B \arrow[d, tail] \\ A \arrow[r, tail] & C \end{tikzcd}$$

Choosing C with $A \cup B = C$, a positive pre-logos has coproducts.

1.624 In a positive pre-logos any $f : A \to B_1 + B_2$ yields a decomposition, $A_i = f^{\#}(B_i)$:

$$\begin{tikzcd} A \arrow[r, "\sim"] \arrow[dr, "f"'] & A_1 + A_2 \arrow[d, "f_1 + f_2"] \\ & B_1 + B_2 \end{tikzcd}$$

1.625 For $\boldsymbol{A}, \boldsymbol{B}$ positive pre-logoi and $T : \boldsymbol{A} \to \boldsymbol{B}$ a representation of regular categories: *T is a representation of pre-logoi iff it preserves disjoint unions.*

1.63 $\Sigma : \boldsymbol{A}/B \to \boldsymbol{A}$ yields $\mathcal{S}ub_{\boldsymbol{A}/B}(-) \simeq \mathcal{S}ub_{\boldsymbol{A}}(\Sigma(-))$, respecting inverse images when $\boldsymbol{A}$ has pullbacks, and unions in $\boldsymbol{A}$ give unions in $\boldsymbol{A}/B$.

If $\boldsymbol{A}$ is a (positive) pre-logos so is $\boldsymbol{A}/B$, and $\Delta : \boldsymbol{A} \to \boldsymbol{A}/B$ is a representation of pre-logoi.

The capitalization lemma [1.543] applies: *any (positive) pre-logos is faithfully representable in a capital (positive) pre-logos.*

1.631 $A_1 \subset A$ is a **COMPLEMENTED SUBOBJECT** if some $A_2 \subset A$ has $A_1 \cap A_2 = 0$, $A_1 \cup A_2 = A$ [1.621]; distributivity of $\mathcal{S}ub(A)$ makes A_2 unique.

In a positive pre-logos a complemented subobject of a projective object is projective. With [1.525]: *the* **COMPLEMENTED SUBTERMINATORS** *— complemented subobjects of* 1 *— in a capital pre-logos are projective.*

1.632 $\mathcal{G} \subset |\boldsymbol{A}|$ is a **GENERATING SET** if $\{(G, -)\}_{G \in \mathcal{G}}$ collectively yields an embedding; $\mathcal{B} \subset |\boldsymbol{A}|$ is a **BASIS** if $\{(B, -)\}_{B \in \mathcal{B}}$ is collectively faithful. Both elementary.

For cartesian $\boldsymbol{A}$, $\mathcal{B}$ is a basis iff every proper $A' \rightarrowtail A$ admits

$$\begin{tikzcd} B' \arrow[r, tail] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & B \arrow[d] \\ A' \arrow[r, tail] & A \end{tikzcd}$$

with $B \in \mathcal{B}$ and $B' \rightarrowtail B$ proper [1.442].

1.633 *A positive pre-logos is capital iff the complemented subterminators are projective and form a basis.*

1.634 $\mathcal{F} \subset P$ is a **PRE-FILTER** if non-empty and any $x, y \in \mathcal{F}$ have $z \in \mathcal{F}$ with $z \leq x$, $z \leq y$; a **FILTER** if also an updeal.

For $\mathcal{F}$ a pre-filter in $\mathcal{V}al_{\boldsymbol{A}}$: $T_{\mathcal{F}}(A)$ has elements named by $x : U \to A$, $U \in \mathcal{F}$, with x and $y : V \to A$ naming the same element when

$$\begin{tikzcd} W \arrow[r] \arrow[d] & U \arrow[d, "x"] \\ V \arrow[r, "y"'] & A \end{tikzcd}$$

for some $W \in \mathcal{F}$, $W \subset U$, $W \subset V$.

$T_{\mathcal{F}}$ preserves finite products and equalizers; it preserves covers when the elements of $\mathcal{F}$ are projective.

In a pre-logos $T_{\mathcal{F}}$ *preserves disjoint unions iff* $0 \notin \mathcal{F}$*, and* $U_1 + U_2 \in \mathcal{F}$ *implies* $U_1 \in \mathcal{F}$ *or* $U_2 \in \mathcal{F}$*.*

1.635 **THE REPRESENTATION THEOREM FOR PRE-LOGOI** *Every small positive pre-logos is faithfully representable in a power of the category of sets.* 'Positive' is removable, every pre-logos being faithfully representable in a positive one [2.217].

In a capital positive pre-logos a representative set $\mathcal{B}$ of complemented subterminators, as a sublattice of $\mathcal{S}ub$, is distributive with every element complemented — a **BOOLEAN ALGEBRA**.

Every pre-filter $\mathcal{F} \subset \mathcal{B}$ extends, by the axiom of choice, to an **ULTRA-FILTER** $\overline{\mathcal{F}}$: a maximal pre-filter with $0 \notin \overline{\mathcal{F}}$, necessarily a filter.

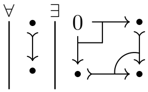
For $\overline{\mathcal{F}}$ an ultra-filter, $T_{\overline{\mathcal{F}}}$ is a representation of pre-logoi. Collectively faithful, because $\mathcal{B}$ is a basis.

1.636 *Any Horn sentence in the predicates of pre-logoi true for the category of sets is true for all positive pre-logoi* — 'positive' again removable [2.217].

1.637 A *special* pre-logos satisfies every universal sentence in the predicates of pre-logoi satisfied by $\mathcal{S}$. A pre-logos is a special regular category iff two-valued.

A pre-logos is special iff $(A' \times B) \cup (A \times B')$ is a proper subobject of $A \times B$ for every pair of proper $A' \subset A$, $B' \subset B$.

1.64 A **BOOLEAN PRE-LOGOS**: a pre-logos in which $\mathcal{S}ub(A)$ is a boolean algebra for all A. $\mathcal{S}ub(A)$ being distributive, only complements need be required, and they are then unique:



1.641 *A representation of pre-logoi between boolean pre-logoi preserves the boolean structure automatically*: each $\mathcal{S}ub(A) \to \mathcal{S}ub(TA)$ preserves complements and is a map of boolean algebras.

1.65 A **PRE-TOPOS**: an effective positive pre-logos. Every pre-logos is faithfully represented in a pre-topos [2.217].

1.651 **THE AMALGAMATION LEMMA** *Monics with a common source $x : A \rightarrowtail B$, $y : A \rightarrowtail C$ in a pre-topos extend to*

$$\begin{tikzcd} A \arrow[r, tail, "y"] \arrow[d, tail, "x"'] & C \arrow[d, tail, "\overline{x}"] \\ B \arrow[r, tail, "\overline{y}"'] & D \end{tikzcd}$$

— take D with level $E = l^\circ x^\circ yr \cup 1 \cup r^\circ y^\circ xl$ on $B + C$, for l, r the coproduct injections. The square is a pullback, hence by [1.62] a pushout.

1.652 *In a pre-topos covers coincide with epics and monics coincide with cocovers* — every monic is an equalizer, so the image of an epic is entire.

1.653 *For $A \to B$ and monic $y : A \rightarrowtail C$ in a pre-topos there exist*

$$\begin{tikzcd} A \arrow[r, tail, "y"] \arrow[d] & C \arrow[d] \\ B \arrow[r, tail] & D \end{tikzcd}$$

— factor $A \to B$ through its image and stack the square with [1.651].

1.654 A pre-topos need not be cocartesian. If it is, the last two sections show its opposite category is regular.

1.655 *If $\boldsymbol{A}, \boldsymbol{B}$ are pre-topoi and $T : \boldsymbol{A} \to \boldsymbol{B}$ preserves* 0*, pushouts, finite products and monics, then T is a bicartesian representation.*

1.657 In an effective regular category, for R tabulated by

$$\begin{tikzcd} & T \arrow[dl, "x"'] \arrow[dr, "y"] & \\ A & & A \end{tikzcd}$$

and $f : A \to B$ a coequalizer of x, y: ff° is the minimal equivalence relation containing R.

A pre-topos is cocartesian iff for every endo-relation R there is a minimal equivalence relation containing R. A functor between bicartesian pre-topoi preserves the bicartesian structure iff it is a representation of pre-logoi preserving minimal equivalence relations.

1.658 An object A in a pre-logos is **DECIDABLE** if the diagonal $\langle 1, 1\rangle : A \rightarrowtail A \times A$ has a complement. *Every object of a pre-topos is decidable iff the pre-topos is boolean* — pullbacks of complemented subobjects are complemented.

1.66 Choice objects [1.57]: *in any regular category a subobject of a choice object is choice, and a quotient of a choice object is choice* — a quotient $x : A \to B$

[1.568] is also a subobject of A, via a map contained in x°.

1.661 *In a regular category finite products of choice objects are choice* — an entire relation targeted at a terminator is already a map.

1.662 *In a pre-topos: (1) binary coproducts of choice objects are choice; (2)* $1+1$ *is choice; (3) the pre-topos is boolean — are equivalent.*

(2) restated by $\mathscr{Rel}(B, 1+1) \simeq \mathscr{Sub}(B) \times \mathscr{Sub}(B)$: an entire relation $B \to 1+1$ is a pair of subobjects covering B, a map $B \to 1+1$ a partition of B — so $X \cup Y = B$ in $\mathscr{Sub}(B)$ gives $X' \subset X$, $Y' \subset Y$ with $X' \cup Y' = B$, $X' \cap Y' = 0$.

1.7 Logoi

1.7 a **LOGOS** is a regular category in which every $\mathcal{Sub}(A)$ is a lattice and every $f^\# : \mathcal{Sub}(B) \to \mathcal{Sub}(A)$ has a right adjoint $f^{\#\#} : \mathcal{Sub}(A) \to \mathcal{Sub}(B)$: $f^\#(B') \subset A'$ iff $B' \subset f^{\#\#}(A')$.

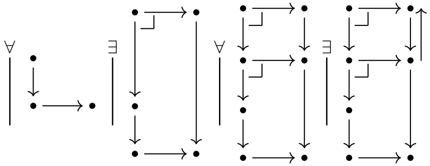
restated: $f^{\#\#}(A')$ is the maximal subobject $B' \subset B$ with $f^\#(B') \subset A'$.

1.71 in $\mathcal{S}$, $f^{\#\#}(A') = \{b \in B \mid \forall_a[f(a) = b \Rightarrow a \in A']\}$. In a boolean pre-logos $f^{\#\#}(A') = \neg(f(\neg A'))$, since $f^\#$ preserves complements.

1.711 a logos is a pre-logos: $f^\#$ preserves every union that exists, since $f^\#(\cup B_i) \subset A'$ iff every $f^\#(B_i) \subset A'$.

1.712 **LOCALLY COMPLETE**: every $\mathcal{Sub}(A)$ is a complete lattice. A locally complete regular category whose $f^\#$ preserve all unions is a logos, with $f^{\#\#}(A') = \cup \{B_i \mid f^\#(B_i) \subset A'\}$.

1.714 the axiom drawn — the corner marks make the squares pullbacks, so each left vertex is the inverse image of the right one; the last panel returns $B'' \subset B'$, so no B'' properly containing B' has $f^\#(B'') \subset A'$.



1.733 A in a pre-logos is **COPRIME** if $(A, \_)$ preserves finite unions: any finite collection of subobjects of A whose union is A contains A. A is **CONNECTED** if it has exactly two complemented subobjects.

1.77 for a binary endo-relation R in a regular category, the **TRANSITIVE CLOSURE** $R^\dagger$ is the minimal transitive relation containing R, and the **TRANSITIVE-REFLEXIVE CLOSURE** R^* the minimal transitive-reflexive one. In a pre-logos each exists iff the other does: $R^* = 1 \cup R^\dagger$ and $R^\dagger = RR^*$.

a **TRANSITIVE LOGOS**, likewise a **TRANSITIVE PRE-LOGOS**, is one in which every endo-relation has a transitive closure.

1.771 countable unions in every $\mathcal{Sub}$, preserved by inverse images (as a logos forces): $R^\dagger = R \cup R^2 \cup \cdots \cup R^n \cup \cdots$, transitive because composition then distributes with countable union.

1.772 a **Σ-TRANSITIVE LOGOS** is a transitive logos in which $R^\dagger$ is the least upper bound of the finite powers of R; a **Σ-TRANSITIVE PRE-LOGOS** wants that union stable: $R^\dagger = \bigcup_{n=1}^{\infty} R^n$, preserved under inverse images.

σ-transitivity is an infinitary Horn sentence: $(R \subset T) \wedge (R^2 \subset T) \wedge \cdots$ implies $R^\dagger \subset T$; for pre-logoi, $f^\#(R^n) \subset A'$ for all n implies $f^\#\left(R^\dagger\right) \subset A'$. Its models are closed under products and substructures, and $\mathcal{S}$ is σ-transitive, so representing transitive pre-logoi in a power of $\mathcal{S}$ requires it.

1.773 *every locally complete regular category is transitive* ($R^\dagger$ is the intersection of all transitive relations containing R), but need not be σ-transitive: $R_1 = R$, $R_{\alpha+1} = R_\alpha \cup (R_\alpha)^2$, $R_\beta = \bigcup_{\alpha<\beta} R_\alpha$ at limits.

1.775 the **EQUIVALENCE CLOSURE** $R^\equiv$ is the minimal equivalence relation containing R; in a transitive pre-logos $R^\equiv = (R \cup R^\circ)^*$. An effective regular category has all equivalence closures iff it has coequalizers.

a pre-logos is **E-STANDARD** if every R has an equivalence closure and $R^\equiv$ is always the stable union of the finite powers of $R^\circ \cup 1 \cup R$. σ-transitive implies E-standard, and E-standardness is necessary for a faithful family of set-valued representations preserving equivalence closures.

1.78 for relations R, S with a common target, R/S is the maximum relation T, if it exists, with $TS \subset R$: $T \subset R/S$ iff $TS \subset R$. The triangle only semi-commutes, as the containment sign says.

$$\begin{tikzcd} \bullet \arrow[dr, "R"] \arrow[dd, "R/S"'] & \\ {} \arrow[r, phantom, "\subset"] & \bullet \\ \bullet \arrow[ur, "S"'] & \end{tikzcd}$$

1.781 **Rel** $x(R/S)y$ iff $\forall_z (ySz) \Rightarrow (xRz)$.

1.782 for a map f with the same target as R: R/f exists and equals Rf°.

1.783 $(R/S_1)/S_2 = R/(S_2S_1)$: if the left side exists so does the right, and they agree.

1.784 *in a logos R/S exists for every pair of relations with a common target.* Every S is $l^\circ r$, reducing the question to R/f°; there $Tf^\circ = (1 \times f)^{\#}(T)$, so $R/f^\circ = (1 \times f)^{\#\#}(R)$.

1.785 conversely, read $A' \rightarrowtail A$ followed by the diagonal $\langle 1, 1\rangle : A \to A \times A$ as a relation on A: if A'/f° exists then $f^{\#\#}(A') = 1_B \cap (A'/f^\circ)^\circ(A'/f^\circ)$.

1.786 for R, S, T with a common target, $(R/S)(S/T) \subset R/T$; and R/R is transitive and reflexive.

1.787 $\overline{R}$ is the minimum reflexive relation with $R\overline{R} \subset \overline{R}$. *in a logos $\overline{R} = R^*$: if either exists so does the other, and they are equal.*

1.8 Adjoint functors, Grothendieck topoi, and exponential categories

1.81 **ADJOINT PAIR OF FUNCTORS** $F : \boldsymbol{A} \to \boldsymbol{B}$, $G : \boldsymbol{B} \to \boldsymbol{A}$: some natural equivalence $(FA, B) \simeq (A, GB)$ exists, natural in both variables — as functors $\boldsymbol{A}^\circ \times \boldsymbol{B} \to \mathcal{S}$. F a **LEFT-ADJOINT** of G, G a **RIGHT-ADJOINT** of F.

$F \dashv G$	$(FA, B) \simeq (A, GB)$
in posets	$FA \leq B$ iff $A \leq GB$
$\boldsymbol{A}' \subset \boldsymbol{A}$ reflective	$\left(\overline{A}, B\right) \simeq (A, B)$, $B \in \boldsymbol{A}'$
$\boldsymbol{A}' \subset \boldsymbol{A}$ coreflective	$(B, GA) \simeq (B, A)$, $B \in \boldsymbol{A}'$
adjoint on the right	$(B, FA) \simeq (A, GB)$, F, G contravariant
adjoint on the left	$(FA, B) \simeq (GB, A)$, F, G contravariant

1.811 poset hom-sets have at most one element, so adjointness is $(FA, B) \neq \emptyset$ iff $(A, GB) \neq \emptyset$. Direct image is a left-adjoint of inverse image $f^{\#}$.

1.813 **REFLECTIVE SUBCATEGORY** $\boldsymbol{A}' \subset \boldsymbol{A}$: the inclusion has a left-adjoint, whose values are **REFLECTIONS**. Elementary form — every A has $A \to \overline{A}$, $\overline{A} \in \boldsymbol{A}'$, and every $A \to B$, $B \in \boldsymbol{A}'$, factors by a unique $\left(\overline{A} \to B\right) \in \boldsymbol{A}'$:

$$\begin{tikzcd} & \overline{A} \arrow[d, dashed] \\ A \arrow[ur] \arrow[r] & B \end{tikzcd}$$

one $A \to \overline{A}$ chosen per A makes reflection a functor: $\overline{x}$ is the unique map with

$$\begin{tikzcd} A_1 \arrow[r] \arrow[d, "x"'] & \overline{A}_1 \arrow[d, dashed, "\overline{x}"'] \\ A_2 \arrow[r] & \overline{A}_2 \end{tikzcd}$$

1.814 most reflective subcategories are full; fullness is necessary and sufficient for the reflection functor to be idempotent.

1.815 **CLOSURE OPERATION**: the reflection functor of a full reflective subcategory P' of a poset P — order-preserving, idempotent, inflationary. $x \le y \Longrightarrow \overline{x} \le \overline{y}$, $\overline{\overline{x}} = \overline{x}$, $x \le \overline{x}$. Conversely its values form a full reflective subcategory.

1.816 **COREFLECTIVE** $\boldsymbol{A}' \subset \boldsymbol{A}$: the inclusion has a right-adjoint.

1.817 G has a left-adjoint F iff every $(A, G(-))$ is representable, by FA; Fx is then the unique map with

$$\begin{tikzcd}
(A_2, G(-)) \arrow[r, "\simeq"] \arrow[d, "{(x, G(-))}"'] & (FA_2, -) \arrow[d, dashed, "{(Fx, -)}"'] \\
(A_1, G(-)) \arrow[r, "\simeq"] & (FA_1, -)
\end{tikzcd}$$

1.818 contravariant $F : \boldsymbol{A} \to \boldsymbol{B}$, $G : \boldsymbol{B} \to \boldsymbol{A}$ are **ADJOINT ON THE RIGHT** if $(B, FA) \simeq (A, GB)$ naturally, **ADJOINT ON THE LEFT** if $(FA, B) \simeq (GB, A)$. Between posets, adjoint on the right = Galois connection.

adjoint on the right iff F then $\boldsymbol{B} \to \boldsymbol{B}^\circ$ is left-adjoint of $\boldsymbol{B}^\circ \to \boldsymbol{B}$ then G; adjoint on the left iff $\boldsymbol{A}^\circ \to \boldsymbol{A}$ then F is left-adjoint of G then $\boldsymbol{A} \to \boldsymbol{A}^\circ$. Covariant theorems thus yield contravariant ones.

1.82 $\boldsymbol{B}^{\boldsymbol{D}}$: objects the functors $\boldsymbol{D} \to \boldsymbol{B}$, maps the natural transformations; $\boldsymbol{D}$ usually small. **DIAGONAL FUNCTOR** $\Delta : \boldsymbol{B} \to \boldsymbol{B}^{\boldsymbol{D}}$: $B \mapsto$ the functor constant at B, $x \mapsto$ the transformation constant at x.

$\boldsymbol{D} \neq \emptyset$: Δ is an embedding, $\boldsymbol{B}$ the subcategory of constant functors. $\boldsymbol{D} = \emptyset$: $\boldsymbol{B}^{\boldsymbol{D}}$ is the terminator among categories, and Δ has a right-adjoint iff $\boldsymbol{B}$ has a terminator, a left-adjoint iff $\boldsymbol{B}$ has a coterminator.

1.821 objects of $\boldsymbol{B}^{\boldsymbol{D}}$ are diagrams modelled on $\boldsymbol{D}$, whose objects i, j, k are the vertices. A map $\Delta(B) \to D$ is a *lower-bound*: a family $\{B \to D_i\}_{|\boldsymbol{D}|}$ compatible with every $x : i \to j$ of $\boldsymbol{D}$.

$$\begin{tikzcd} & D_i \arrow[d, "D_x"] \\ B \arrow[ur] \arrow[dr] & \\ & D_j \end{tikzcd}$$

D has a coreflection in Δ iff it has a greatest lower-bound $\{L \to D_i\}$: every lower-bound has a unique map over it, all i.

$$\begin{tikzcd} B \arrow[rr, dashed] \arrow[dr] & & L \arrow[dl] \\ & D_i & \end{tikzcd}$$

Discrete $\boldsymbol{D}$: compatibility evaporates, greatest lower-bound = product diagram.

1.822 **LIMITS** $\lim D$: the values of a right-adjoint of $\Delta : \boldsymbol{B} \to \boldsymbol{B}^{\boldsymbol{D}}$. **COLIMITS** colim D: the values of a left-adjoint — least upper bounds.

$\lim \emptyset$ is a terminator, colim $\emptyset$ a coterminator. For $\boldsymbol{D}$ finitely presented by $\cdot \rightrightarrows \cdot$ (two objects, two non-identity maps) the right-adjoint delivers equalizers, the left-adjoint coequalizers.

1.823 **COMPLETE**: every small diagram has a limit. **COCOMPLETE**: every small diagram has a colimit.

1.824 a family of monics $\{A_i \rightarrowtail A\}_I$ is a diagram modelled on I' (I discrete plus a terminator). Its limit is more than the intersection in $\mathcal{Sub}(A)$: any $B \to A$, monic or not, factors through $\lim A_i$ iff it factors through every A_i.

1.825 *complete iff equalizers and arbitrary products* dually *cocomplete iff coequalizers and arbitrary coproducts* and *(co-)cartesian iff every finite diagram has a (co-)limit.*

1.827 **CONTINUOUS**: preserves all small limits — from a complete source, iff it preserves equalizers and arbitrary products. **COCONTINUOUS**: preserves all small colimits, from a cocomplete source iff it preserves coequalizers and arbitrary coproducts. A contravariant functor is continuous if it carries colimits to limits.

1.828 *weak*- removes uniqueness. **WEAK-LIMIT** of D: a compatible $\{W \to D_i\}$ admitting, for every compatible $\{B \to D_i\}$, at least one

$$\begin{tikzcd} B \arrow[rr, dashed] \arrow[dr] & & W \arrow[dl] \\ & D_i & \end{tikzcd}$$

WEAKLY-COMPLETE: every small diagram has a weak-limit.

a functor preserving one weak-limit of a diagram preserves them all; a continuous functor from a complete category preserves all weak-limits.

1.829 avoid "weakly continuous": *a functor that preserves weak-limits preserves limits* — preserving weak-limits already forces preservation of monic families.

1.82(10) **PRE-LIMIT** for $D : \boldsymbol{D} \to \boldsymbol{B}$: a *set* of lower-bounds cofinal in the class of all of them — compatible families $\{\{P_j \to D_i\}_{|\boldsymbol{D}|}\}_J$ such that every lower-bound $\{B \to D_i\}$ admits $j \in J$ with

$$\begin{tikzcd} B \arrow[rr, dashed] \arrow[dr] & & P_j \arrow[dl] \\ & D_i & \end{tikzcd}$$

for all $i \in |\boldsymbol{D}|$.

PRE-COMPLETE: every small diagram has a pre-limit. Preserving pre-limits $\implies$ preserving weak-limits $\implies$ [1.829] preserving limits.

1.83 **PRE-ADJOINT** for $A \in \boldsymbol{A}$, given $G : \boldsymbol{B} \to \boldsymbol{A}$: a set $\{A \to G(B_i)\}_I$ such that every $A \to G(B)$ admits $i \in I$, $x : B_i \to B$ with

$$\begin{tikzcd} & A \arrow[dl] \arrow[dr] & \\ G(B_i) \arrow[rr, dashed, "G(x)"'] & & G(B) \end{tikzcd}$$

Locally small $\boldsymbol{A}$: a set $\{B_i\}_I$ in $\boldsymbol{B}$ through one of which every $A \to G(B)$ factors this way.

$\boldsymbol{B}$ full, G its inclusion: a pre-adjoint is a **PRE-REFLECTION** of A — a set $\{B_i\}$ through one of which every map from A to a $\boldsymbol{B}$-object factors. **PRE-ADJOINT FUNCTOR**: every $A \in \boldsymbol{A}$ has a pre-adjoint.

THE GENERAL ADJOINT FUNCTOR THEOREM *If $\boldsymbol{B}$ is locally small and complete then $G : \boldsymbol{B} \to \boldsymbol{A}$ has a left-adjoint iff it is continuous and pre-adjoint.*

1.831 **UNIFORMLY CONTINUOUS** $G : \boldsymbol{B} \to \boldsymbol{A}$: for every small $D : \boldsymbol{D} \to \boldsymbol{B}$, G carries the lower-bounds of D to a cofinal family of lower-bounds of $\boldsymbol{D} \overset{D}{\to} \boldsymbol{B} \overset{G}{\to} \boldsymbol{A}$ — every $\{A \to G(D_i)\}$ admits a lower-bound $\{B \to D_i\}$ and

$$\begin{tikzcd} A \arrow[rr, dashed] \arrow[dr] & & G(B) \arrow[dl] \\ & G(D_i) & \end{tikzcd}$$

for all i.

such G preserves pre-limits, hence weak-limits, hence limits. $\boldsymbol{B}$ pre-complete: uniformly continuous = preserves pre-limits. $\boldsymbol{B}$ (weakly) complete: = preserves (weak) limits.

THE MORE GENERAL ADJOINT FUNCTOR THEOREM *If $\boldsymbol{B}$ is locally small and idempotents split in $\boldsymbol{B}$ then G has a left-adjoint iff it is uniformly continuous and pre-adjoint.*

1.832 **POINTWISE CONTINUOUS** $T : \boldsymbol{B} \to \mathcal{S}$: for every small D, every lower-bound $\{1 \to T(D_i)\}$ in $\mathcal{S}$ comes from a lower-bound $\{B \to D_i\}$ in $\boldsymbol{B}$ and a point

$$\begin{tikzcd} 1 \arrow[rr, dashed] \arrow[dr] & & T(B) \arrow[dl] \\ & T(D_i) & \end{tikzcd}$$

G is uniformly continuous iff every $(A, G(-))$ is pointwise continuous.

1.833 smallest subfunctor $T' \subset T$ containing $\{x_i \in T(B_i)\}_I$: $y \in T'(B)$ iff $(Tf)(x_i) = y$ for some $i \in I$, $f : B_i \to B$. $T' = T$: T is generated by the x_i. **PETTY-FUNCTOR**: generated by some set of elements. G is pre-adjoint iff every $(A, G(-))$ is petty.

1.834 **THE GENERAL REPRESENTABILITY THEOREM** *If $\boldsymbol{B}$ is locally small and idempotents split in $\boldsymbol{B}$ then $T : \boldsymbol{B} \to \mathcal{S}$ is representable iff it is a pointwise continuous petty-functor.*

1.835 *A locally small category in which idempotents split has a coterminator iff it has a pre-coterminator and all small diagrams have lower-bounds.*

1.836 $\hat{\boldsymbol{B}} = \mathrm{Split}(\boldsymbol{B})$ formally splits every idempotent; $\boldsymbol{B} \to \hat{\boldsymbol{B}}$ is full, uniformly continuous and pre-adjoint, with a left-adjoint only where idempotents already split. *A set-valued functor from a locally small category is a retract of a representable iff it is a pointwise continuous petty-functor.*

1.837 $\boldsymbol{B}$ complete: $\Delta : \boldsymbol{B} \to \boldsymbol{B}^{\boldsymbol{D}}$ is continuous, and pre-adjoint iff all diagrams modelled on $\boldsymbol{D}$ have pre-colimits. Hence *a complete locally small category is cocomplete iff it is pre-cocomplete.*

1.838 **WELL-POWERED** $\boldsymbol{B}$: $\mathcal{Sub}(B)$ is a set for every B. Distinct maps have distinct graphs, so this implies local smallness.

minimal object: no proper subobject (an atom is not minimal — it has exactly one). *Each object of a complete well-powered category has a unique minimal subobject.*

minimal objects form an initial class $\mathcal{M}$, pre-ordered ($f, g : M \to A$ have a non-proper equalizer, so $f = g$), every subset of which has a lower bound; an initial set exists iff $\mathcal{M}$ has one. Any map between minimal objects is a cover.

1.839 pre-adjoints come from a *cardinality function* K: objects to cardinals, with $K^{\#(\alpha)}$ a set of isomorphism types for every cardinal α. For continuous $G : \boldsymbol{B} \to \boldsymbol{A}$, $\boldsymbol{B}$ complete well-powered: B is *generated by* A *through* G if $A \to G(B)$ factors through no $G(B') \subset G(B)$ with $B' \subsetneq B$.

1.83(10) **COGENERATING SET** $\{C_i\}_I$: the $\{(-, C_i)\}_I$ are collectively an embedding — $f \neq g : A \to B$ gives $i \in I$, $h : B \to C_i$ with $fh \neq gh$. In a complete category: iff every object is embeddable in a product of the C_i.

THE SPECIAL ADJOINT FUNCTOR THEOREM *If* $\boldsymbol{B}$ *is complete, well-powered and has a cogenerating set, and* $\boldsymbol{A}$ *is locally small, then every continuous* $G : \boldsymbol{B} \to \boldsymbol{A}$ *has a left-adjoint.*

A complete well-powered category with a cogenerating set has a coterminator — the minimal subobject of $\Pi_I C_i$.

1.83(11) dually: *if* $\boldsymbol{A}$ *is cocomplete, well-co-powered and has a generating set, every cocontinuous functor from* $\boldsymbol{A}$ *to a locally small category has a right-adjoint, and every contravariant continuous functor to a locally small category has an adjoint on the right.*

1.84 **GIRAUD DEFINITION** of a Grothendieck topos (the first of four): locally small, cocomplete, effective regular, pullbacks preserve arbitrary unions, disjoint unions of objects exist, and there is a generating set. Such is a pre-topos and a logos.

1.842 *If* $\boldsymbol{E}$ *is a Grothendieck topos then the graphing functor* $\boldsymbol{E} \to \mathcal{Rel}(\boldsymbol{E})$ *has a right-adjoint.* Via 1.843–6 and the special adjoint functor theorem; chapter 2 proves it directly.

1.843 *A Grothendieck topos is well-powered*: a generating set $\{G_i\}_I$ is a basis, every subobject being an equalizer, so $B' \subset B$ is fixed by $\{(G_i, B') \subset (G_i, B)\}_I$. Hence $\mathcal{Rel}(\boldsymbol{E})$ *is locally small.*

comonics coincide with covers, and covers out of A correspond to the equivalence relations on A: *a Grothendieck topos is well-copowered.*

1.844 *A Grothendieck topos is locally complete* — the union of $\{A_i \rightarrowtail A\}$ is the image of $\Sigma A_i \to A$, Σ the disjoint union. Inverse images preserve arbitrary unions, so *composition of relations distributes with arbitrary unions.*

1.845 coproduct $\{u_i : A_i \to S\}_I$: $u_i u_i^\circ = 1$, $u_i u_j^\circ = 0$ for $i \neq j$, $\bigcup u_i^\circ u_i = 1_S$ — arbitrary coproducts are disjoint unions. *A coproduct in* $\boldsymbol{E}$ *remains a coproduct in* $\mathcal{Rel}(\boldsymbol{E})$: $R = \bigcup u_i^\circ R_i$ is the unique relation with $u_i R = R_i$, all i.

1.846 *A coequalizer in* $\boldsymbol{E}$ *remains a coequalizer in* $\mathcal{Rel}(\boldsymbol{E})$. A Grothendieck topos is $\boldsymbol{E}$-standard: the smallest equivalence relation containing R is $\bigcup_{n=-\infty}^{\infty} R^n$, with $R^0 = 1$, $R^{-n} = (R^\circ)^n$.

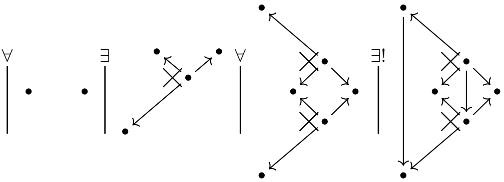
1.847 **Rel** the ordering relation on the two-element ordered set is idempotent in $\mathcal{Rel}(\mathcal{S})$ but cannot be split. So $\mathcal{Rel}(\boldsymbol{E})$ is not cocomplete, though disjoint unions give it coproducts — and, reciprocation making it self-dual, products too.

1.85 **EXPONENTIAL** $\boldsymbol{A}$ with binary products (elsewhere: cartesian closed): every $A \times -$ has a right-adjoint; equivalently every $(A \times -, B)$ is representable, by an object B^A.

an equivalence $\eta : (-, B^A) \to (A \times -, B)$ carries $1 \in \left(B^A, B^A\right)$ to an **EVALUATION MAP** $e : A \times B^A \to B$, and every $f : A \times X \to B$ has a unique $\lambda f : X \to B^A$ with

$$\begin{tikzcd}
& A \times X \arrow[dl, dashed, "1 \times \lambda f"'] \arrow[dr, "f"] & \\
A \times B^A \arrow[rr, "e"'] & & B
\end{tikzcd}$$

in the diagrammatic language — the cross marking a product diagram:



1.852 a poset is exponential iff it has binary intersections and every a, b have b^a with $x \leq b^a$ iff $a \wedge x \leq b$.

1.853 B^A is a bifunctor $\boldsymbol{A}^\circ \times \boldsymbol{A} \to \boldsymbol{A}$, with $f^A = \lambda\left(A \times B_1^A \xrightarrow{e} B_1 \xrightarrow{f} B_2\right)$ and $B^g = \lambda\left(A_1 \times B^{A_2} \xrightarrow{g \times 1} A_2 \times B^{A_2} \xrightarrow{e} B\right)$:

$$\begin{tikzcd}
B_1^{A_2} \arrow[r, "f^{A_2}"] \arrow[d, "B_1^g"'] & B_2^{A_2} \arrow[d, "B_2^g"] \\
B_1^{A_1} \arrow[r, "f^{A_1}"'] & B_2^{A_1}
\end{tikzcd}$$

$(-)^A$ has the left-adjoint $A \times -$, so preserves limits. Commutativity of products makes $B^{(-)}$ adjoint on the right to itself, $\left(A_1, B^{A_2}\right) \simeq (A_2 \times A_1, B) \simeq \left(A_2, B^{A_1}\right)$, hence continuous: colimits to limits.

bicartesian exponential: $1^A \simeq 1$, $(B_1 \times B_2)^A \simeq B_1^A \times B_2^A$, $B^0 \simeq 1$, $B^{A_1+A_2} \simeq B^{A_1} \times B^{A_2}$, $\left(B^{A_1}\right)^{A_2} \simeq B^{A_1 \times A_2}$, $B^1 \simeq B$.

$A \times -$ has a right-adjoint, so preserves colimits: $A \times (B_1 + B_2) \simeq (A \times B_1) + (A \times B_2)$, $A \times 0 \simeq 0$ — the last equivalent to strictness of the coterminator.

caution: 0^A is not 0, and $0^0 \simeq 1$. $(-)^A$ preserves monics, so 0^A is a subterminator: the largest U with $A \times U = 0$.

1.854 in any category with binary products, $\Sigma : \boldsymbol{A}/B \to \boldsymbol{A}$ is a left-adjoint of $\Delta : \boldsymbol{A} \to \boldsymbol{A}/B$: $(\Sigma f, C) \simeq (f, \Delta C)$, with $f : A \to B$ a slice object and ΔC the projection $B \times C \to B$.

a right-adjoint of Δ, where it exists, is written $\Pi : \boldsymbol{A}/B \to \boldsymbol{A}$: products of families in $\boldsymbol{A}$ indexed on B.

If $\Pi : \boldsymbol{A}/B \to \boldsymbol{A}$ *exists then* A^B *exists for every* A: $(B \times -, A) \simeq (\Sigma\Delta(-), A) \simeq (\Delta(-), \Delta A) \simeq (-, \Pi\Delta(A))$.

If $\boldsymbol{A}$ *is cartesian and exponential then* $\Delta : \boldsymbol{A} \to \boldsymbol{A}/B$ *has a right-adjoint* Π *for every* B — for $f : A \to B$ take the pullback

$$\begin{tikzcd} \Pi(f) \arrow[r, tail] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & A^B \arrow[d, "f^B"] \\ 1 \arrow[r, "\lambda 1_B"'] & B^B \end{tikzcd}$$

$(C, \Pi(f))$, as a subset of $(B \times C, A)$, is then the set of g with

$$\begin{tikzcd} B \times C \arrow[rr, "g"] \arrow[dr, "\ell"'] & & A \arrow[dl, "f"] \\ & B & \end{tikzcd}$$

which is exactly $(\Delta(C), f)$.

1.856 the slice lemma fails for exponential categories: $\boldsymbol{P}$ the posets, 3 the three-element chain, $\boldsymbol{P}/3$ not exponential. $\boldsymbol{P}/A$ *is exponential iff* A *contains no three-element chain.*

1.857 *A full coreflective subcategory of an exponential category closed under binary products is exponential*: $(A \times C, B) \simeq (C, B^A) \simeq (C, G(B^A))$ for the coreflection G.

EXPONENTIAL IDEAL: a full $\boldsymbol{A}' \subset \boldsymbol{A}$ with $B^A \in \boldsymbol{A}'$ for all $A \in \boldsymbol{A}$, $B \in \boldsymbol{A}'$. **REPLETE SUBCATEGORY**: closed under isomorphism type.

A full replete reflective subcategory of an exponential category is an exponential ideal iff reflections preserve products — from $(C, B^A) \simeq (\overline{C}, B^A)$, every $C \to B^A$ with $B \in \boldsymbol{A}'$ extends uniquely:

$$\begin{tikzcd} C \arrow[rr] \arrow[dr] & & \overline{C} \arrow[dl, dashed] \\ & B^A & \end{tikzcd}$$

At $C = B^A$ this forces $C \to \overline{C}$ to be an isomorphism.

1.858 **KURATOWSKI INTERIOR OPERATION** on a lattice: deflationary, idempotent, intersection-preserving — $x^\bullet \leq x$, $x^\bullet = (x^\bullet)^\bullet$, $(x \wedge y)^\bullet = x^\bullet \wedge y^\bullet$. Its values are the *open* elements.

LAWVERE-TIERNEY CLOSURE OPERATION (L-T): inflationary, idempotent, intersection-preserving — $x \leq \overline{x}$, $\overline{\overline{x}} = \overline{x}$, $\overline{\overline{x} \wedge \overline{y}} = \overline{x} \wedge \overline{y}$. Its values are the *closed* elements.

1.859 **BASEABLE** B, in a category with binary products: B^A exists — i.e. $(A \times -, B)$ is representable — for every A. The baseable objects form a full subcategory $\boldsymbol{B}$, with $f^A : B_1^A \to B_2^A$ defined as if all objects were baseable.

The inclusion $\boldsymbol{B} \to \boldsymbol{A}$ *preserves equalizers*: the equalizer of f^A, g^A is B_0^A. $\boldsymbol{B}$ is itself always exponential, since $\left(B^A\right)^{A'} \simeq B^{A \times A'}$.

1.9 Topoi

1.9 Any category with pullbacks composes a map with a relation: for $f : A \to B$ and a table $\langle R; b, c\rangle$ on B, C, pull b back along f.

$$\begin{tikzcd} R' \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & R \arrow[r, "c"] \arrow[d, "b"'] & C \\ A \arrow[r, "f"] & B \end{tikzcd}$$

$R' \to A$, $R' \to C$ is still a monic pair; that table is fR. No regularity needed, and $g(fR) = (gf)R$.

U targeted at C is a **UNIVERSAL RELATION** if every relation R targeted at C is uniquely fU; that f is written ΛR. A **POWER-OBJECT** of C, written $[C]$, is the source of a universal relation, itself written $\ni \subset [C] \times C$.

$$\begin{tikzcd} R \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \ni \arrow[r] \arrow[d] & C \\ A \arrow[r, dashed, "\Lambda R"] & {[C]} \end{tikzcd}$$

A **TOPOS** is a cartesian category in which each object has a power-object. *A topos is a positive effective transitive logos*, hence a pre-topos.

1.911 $\mathscr{Rel}(-, B)$ is contravariant set-valued on any category with pullbacks, $\mathscr{Rel}(f, B)$ sending R to fR. $[B]$ exists iff $\mathscr{Rel}(-, B) \simeq (-, [B])$. A regular $\boldsymbol{A}$ is a topos iff the graphing functor $\boldsymbol{A} \to \mathscr{Rel}(\boldsymbol{A})$ [1.564] has a right adjoint. In $\mathcal{S}$, $[B]$ is the set of subsets of B.

1.912 $\mathscr{Rel}(-, 1) \simeq \mathscr{Sub}(-)$, so $[1]$ — traditionally Ω — is a **SUBOBJECT CLASSIFIER**. A universal subobject is a monic $\Omega' \rightarrowtail \Omega$ of which every monic $A' \rightarrowtail A$ is a unique pullback; χ_A is the **CHARACTERISTIC MAP** of $A' \subset A$, equivalently the unique χ with $\chi^{\#}(\Omega') = A'$.

$$\begin{tikzcd} A' \arrow[r, tail] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & A \arrow[d, dashed, "\chi_A"] \\ \Omega' \arrow[r, tail] & \Omega \end{tikzcd}$$

Uniqueness of χ_A forces uniqueness of $A' \to \Omega'$. Taking $A' \rightarrowtail A$ to be 1_A gives a unique pullback for every A, so each object has exactly one map to Ω': the universal subobject is a point, henceforth written $1 \xrightarrow{t} \Omega$.

1.913 *In any cartesian category with a subobject classifier every subobject is an equalizer* — $A' \rightarrowtail A$ equalizes χ_A and $p_A t$. *Consequently covers coincide with epics.*

1.914 An n-ary $g : \Omega^n \to \Omega$ induces $\hat{g}$ on each $\mathscr{S}ub(A)$: $\hat{g}(A_1, ..., A_n)$ is the subobject with characteristic map $\langle \chi_{A_1}, ..., \chi_{A_n} \rangle g$. Every operation on $\mathscr{S}ub(-)$ so arises, and each commutes with inverse images: $\hat{g}\big(f^{\#}A_1, ..., f^{\#}A_n\big) = f^{\#}\hat{g}(A_1, ..., A_n)$.

$g = \chi$ of $\langle t, t \rangle : 1 \to \Omega \times \Omega$ gives $\hat{g}(A_1, A_2) = A_1 \cap A_2$: Ω is a semi-lattice with unit $1 \xrightarrow{t} \Omega$. $g = \chi$ of $\langle 1, 1 \rangle : \Omega \to \Omega \times \Omega$ makes $\hat{g}(A_1, A_2)$ the equalizer of χ_{A_1}, χ_{A_2}, so $A' \subset \hat{g}(A_1, A_2)$ iff $A_1 \cap A' = A_2 \cap A'$.

1.919 *Every monic endomorphism of Ω is an involution.* Absence of non-automorphic monic endomorphisms is a finiteness condition; Ω need not satisfy the dual condition.

1.91(10) *A nonempty category with binary products, equalizers and power-objects has a terminator, hence is a topos* — the equalizer of $1_{[B]}$ and the constant $\Lambda M_{[B],B}$, where $M_{A,B}$ is the relation tabulated by $A \times B$.

1.92 $\Lambda 1 : B \rightarrowtail [B]$ is the **SINGLETON MAP** in $\mathscr{S}$ it sends b to $\{b\}$. For any $f : A \to B$, $(f\Lambda 1)\ni = f$ and $f\Lambda 1$ is a map, hence $f\Lambda 1 = \Lambda f$; consequently $\Lambda 1$ is monic.

A topos is exponential. Power-objects are baseable, $(A \times -, [B]) \simeq \mathscr{S}ub(- \times A \times B) \simeq (-, [A \times B])$, so $[B]^A$ is constructible as $[A \times B]$; and B is the equalizer of the characteristic map χ of $\Lambda 1$ and of pt, hence baseable [1.859].

$$
\begin{tikzcd}
B \arrow[r, tail, "\Lambda 1"] & {[B]} \arrow[r, bend left, "\chi"] \arrow[r, bend right, "pt"'] & \Omega
\end{tikzcd}
$$

1.921 $\mathscr{R}el(-,B) \simeq \mathscr{S}ub(-\times B) \simeq (-\times B, \Omega) \simeq (-,\Omega^B)$: $[B]$ is definable as Ω^B. Lawvere's original axioms — bicartesian, exponential, with partial map classifiers [1.934] — follow from power-objects alone.

1.922 $[-]$ is the covariant power-object functor, right adjoint of the graphing functor; Ω^- is contravariant. For $g : B_1 \to B_2$, Ω^g is the unique map inducing $R \mapsto Rg^\circ$ from $\mathscr{R}el(-,B_2)$ to $\mathscr{R}el(-,B_1)$; so $\Omega^g = [g^\circ]$.

1.923 B^A is cut out of $[A\times B]$ by a pullback.

$$\begin{tikzcd} B^A \arrow[r, tail] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & {[A\times B]} \arrow[d, "f"'] \\ 1 \arrow[r, "\ulcorner A \urcorner"] & {[A]} \end{tikzcd}$$

In $\mathcal{S}$, f sends $R \subset A\times B$ to $\{a \mid \exists! b, \langle a,b\rangle \in R\}$, and $1 \to [A]$ names the entire subobject.

1.93 *Slice lemma for topoi: for a topos* $\boldsymbol{E}$ *and an object* B, $\boldsymbol{E}/B$ *is a topos and* $\Delta : \boldsymbol{E} \to \boldsymbol{E}/B$ *is a representation of topoi.* $\Delta[A]$ is the power-object of $\Delta(A)$; and for $A' \subset A$, $[A']$ is the equalizer of $1_{[A]}$ and the idempotent $e = \Lambda(\ni \cap ([A]\times A'))$.

1.931 In any cartesian category $f : A \to B$ gives $f^\# : \boldsymbol{A}/B \to \boldsymbol{A}/A$ by pulling back, with left adjoint Σ_f by composing.

$$\begin{tikzcd} f^\# C \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & C \arrow[d] \\ A \arrow[r, "f"] & B \end{tikzcd}$$

THE FUNDAMENTAL LEMMA OF TOPOI *For* $f : A \to B$ *in a topos* $\boldsymbol{E}$, $f^\# : \boldsymbol{E}/B \to \boldsymbol{E}/A$ *has a right adjoint* $\Pi_f : \boldsymbol{E}/A \to \boldsymbol{E}/B$.

1.932 $f^\#$ restricted to $\mathscr{S}ub(B)$ is inverse image, so Π_f restricted to $\mathscr{S}ub(A)$ is the double-sharp operation [1.7]. *The double-sharp axiom holds for topoi.*

1.933 *Pullbacks preserve covering families: for* $f : A \to B$ *and a cover* $\{C_i \to B\}_I$, *the* $\{D_i \to A\}$ *below are a cover.*

$$\begin{tikzcd} D_i \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & C_i \arrow[d] \\ A \arrow[r, "f"] & B \end{tikzcd}$$

Consequently a topos is pre-regular.

1.934 $\mathcal{Par}(\boldsymbol{C})$ has the objects of $\boldsymbol{C}$, and as morphisms $A \to B$ the pairs $\langle A', f\rangle$ with $A' \subset A$, $f : A' \to B$; composition of $\langle A', f\rangle$ and $\langle B', g\rangle$ is $\langle f^{\#}(B'), fg\rangle$.

For a topos $\boldsymbol{E}$ the inclusion $\boldsymbol{E} \subset \mathcal{Par}(\boldsymbol{E})$ has a right adjoint: $\mathcal{Par}(-, A) \simeq \left(-, \tilde{A}\right)$.

$$\begin{tikzcd} B' \arrow[r, tail] \arrow[d, "f"'] \arrow[dr, phantom, "\lrcorner", very near start] & B \arrow[d, dashed] \\ A \arrow[r, tail] & \tilde{A} \end{tikzcd}$$

$\tilde{A} = \Pi_t(A)$; $\tilde{1} = \Omega$, and in a boolean positive logos $\tilde{A} = A + 1$.

1.935 Topoi are pre-regular and satisfy the slice condition [1.541]. *Every topos may be faithfully represented in a capital topos.*

1.94 For $F \subset [A]$, $\Gamma(F)$ is a subset of $\mathcal{Sub}(A)$: the **SUBOBJECTS NAMED BY** F. The **INTERNALLY DEFINED INTERSECTION** $\cap F$:

$$\begin{tikzcd} \cap F \arrow[r, tail] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & A \arrow[d] \\ 1 \arrow[r] & \Omega^F \end{tikzcd}$$

$A \to \Omega^F$ is the adjoint of $F \rightarrowtail [A] \simeq \Omega^A$, and $1 \to \Omega^F$ the adjoint of $F \to 1 \xrightarrow{t} \Omega$.

1.941 *$\cap F$ is preserved by representations of topoi, and is the largest subobject of A that stays a lower bound of the subobjects named by $T(F)$ under every representation T.* "Internally defined" is coextensive with being preserved by representations.

1.942 The adjoint $1 \to \Omega^A$ of $\chi_{A'}$ is the **NAME OF** A', written $\ulcorner A' \urcorner$; A' is named by F iff $\ulcorner A' \urcorner \subset F$, and the $1 \to \Omega^F$ of $\cap F$ is the name of the entire subobject. Ω^f is the internally defined inverse image: $\ulcorner B' \urcorner \Omega^f = \ulcorner f^{\#} B' \urcorner$.

For $A' \subset A$ named by F, the composite $A \to \Omega^F$ followed by $\Omega^{\ulcorner A' \urcorner}$ is $\chi_{A'}$.

1.943 *$\cap F$ is contained in every subobject named by F. If F is well-pointed, $\cap F$ is their greatest lower bound, and $B \to A$ factors through $\cap F$ iff it factors through each subobject named by F.*

1.944 *A topos has a strict coterminator* — $\cap\,\Omega$ in $\mathcal{Sub}(1)$, minimal and stably so under every $p^\#$.

1.945 *A topos is regular.* Image of $f : A \to B$: let $F \subset [B]$ name the B' with $f^\#(B') = A$; then $\cap\, F = \operatorname{Im} f$.

1.946 *A topos is a logos.* Binary unions: $F_i \subset [A]$ names the A' containing A_i; $F = F_1 \cap F_2$ names those containing both, and $\cap\, F = A_1 \cup A_2$.

1.947 *A topos is a transitive logos.* For reflexive R on B: F_1, the equalizer of 1 and $[R \times 1]$ in $[B \times B]$, names the S with $RS \subset S$; F_2 names the reflexive ones; $\cap\,(F_1 \cap F_2) = R^*$.

1.949 The **INTERNALLY DEFINED UNION** $\cup\, F$ is the direct image $\mu(\ni \cap (F \times A))$ along $\mu : [A] \times A \to A$. *It is an upper bound of every subobject named by F, and the least one when F is well-pointed.*

$A' \subset A$ is a permanent lower (upper) bound of F if $T(A')$ stays one under every representation T. $\cap\, F$ *is the greatest permanent lower bound,* $\cup\, F$ *the least permanent upper bound.*

1.94(10) $\Lambda 1 : A \to [A]$ names the subobjects which are points; $\cup\,(\Lambda 1)$ is always entire, and is their least upper bound only when A is well-pointed.

A subobject of A is its **WELL-POINTED PART** $A\cdot$ if it is the least upper bound of the points of A, equivalently the greatest well-pointed subobject; no internal construction of $A\cdot$ is possible. A **SOLVABLE TOPOS** is one in which every object has a well-pointed part; there $\mathcal{Sub}(A)$ has bounds for every elementarily definable family, but the slice lemma fails. Capital topoi are solvable, so every topos is faithfully represented in a solvable one.

1.95 *A topos is a pre-topos.*

1.951 *A topos is effective.* For an equivalence relation E on A, factor ΛE as $A \xrightarrow{h} B \rightarrowtail [A]$; then $h^\circ h = 1$ and $hh^\circ = E$.

1.952 *A topos is positive.*

$$\begin{tikzcd}
0 \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & 1 \arrow[d, "\ulcorner 0 \urcorner"] \\
A \arrow[r, "\Lambda 1"] & {[A]}
\end{tikzcd}$$

since ΛR factors through $\Lambda 1$ iff R is a map, and through $\ulcorner 0 \urcorner$ iff $R = 0$. So $A + 1$ exists; generally $\langle \Lambda 1, \ulcorner 0 \urcorner \rangle$ and $\langle \ulcorner 0 \urcorner, \Lambda 1 \rangle$ are disjoint monics into $[A] \times [B] \simeq [A + B]$, tabulating $A + B$.

1.954–5 *A topos has coequalizers*: for $f, g : A \to B$ put $R = f^\circ g$ and $S = (R \cup R^\circ)^*$; effectiveness makes S the level of some $B \to C$, the coequalizer. *A topos is bicartesian* (coequalizers, coterminator, binary coproducts).

1.961 E is an **INJECTIVE** if $(-, E)$ carries monics to epics.

$$\begin{tikzcd} A \arrow[r, tail] \arrow[d] & B \arrow[dl, dashed] \\ E & \end{tikzcd}$$

In a pre-topos, pushouts of monics are monic [1.651], so E is injective iff every monic $E \rightarrowtail A$ has a right inverse.

E in an exponential category is **INTERNALLY INJECTIVE** if E^- carries monics to epics. *In a topos Ω is internally injective*: for monic f, $\Omega^f = [f^\circ]$ has left inverse $[f]$, since $\Omega^f[f] = [f^\circ f] = 1$. *In a topos Ω is (externally) injective.*

1.962 *If E is injective in an exponential category then so is E^A*, since $(-, E^A) \simeq (- \times A, E)$ and $- \times A$ preserves monics; likewise internally. Hence Ω^A is injective, and $\Lambda 1$ embeds every object in an injective.

1.963 *$\tilde{A}$ [1.934] is injective* — a partial map $B_1 \to A$ is a partial map $B_2 \to A$ whenever $B_1 \rightarrowtail B_2$.

1.964 A category is **VALUE-BASED** if its values form a basis; capital topoi are. *In a value-based topos Ω is a cogenerator.*

1.965 C in an exponential category **INTERNALLY COGENERATES** if C^- is a contravariant embedding; a cogenerator does. *In a topos Ω internally cogenerates.*

1.966 A **PROGENITOR** is an object whose subobjects form a generating set. A topos is value-based iff 1 is a progenitor; every Grothendieck topos has one. *If G is a progenitor then Ω^G is a cogenerator.*

1.967 *Equivalent on a locally small topos: (a) arbitrary powers exist; (b) arbitrary copowers exist; (c) arbitrary copowers of* 1 *exist. Each implies local completeness.* (a) says contravariant representables have adjoints, (b) that covariant ones do, (c) that Γ does.

1.968 *A locally small topos is complete iff it is cocomplete.*

1.969 The **LAWVERE DEFINITION** of a Grothendieck topos: a cocomplete topos with a generating set. The **TIERNEY DEFINITION**: a topos with a progenitor and arbitrary copowers of 1 — equivalently, with a geometric morphism to $\mathcal{S}$.

1.96(11) *Slice lemma for Grothendieck topoi: $\boldsymbol{E}/B$ is a Grothendieck topos, and $\Delta : \boldsymbol{E} \to \boldsymbol{E}/B$ preserves progenitors.*

1.971 A is small if it is a subquotient of a $\boldsymbol{G}$-object: $A \leftarrow G' \rightarrowtail G$ with the left arrow a cover. *In a boolean Grothendieck topos every object is a coproduct of small objects.*

1.972 *In a boolean logos with* 1 *projective,* 1 *is a progenitor. Equivalent on a Grothendieck topos: (1) boolean and* 1 *projective; (2) boolean and value-based; (3) boolean with a choice progenitor; (4) every object a coproduct of sub-terminators; (5) AC.*

1.973 A topos is **IAC** (internal axiom of choice) if $(-)^A$ [1.853] preserves epics for every A.

1.974 *A topos is AC iff it is IAC and* 1 *is projective.* Every functor from an AC topos preserves epics, since it preserves left-invertibility.

1.976 *A small topos may be faithfully represented in an AC topos iff it is IAC* — take $C_A \subset A^{[A]^+}$ naming the choice maps contained in the universal relation.

1.977 *Any universal sentence in the predicates of topoi which follows from AC follows from IAC.* In particular *an IAC topos is boolean.*

1.979 *For any topos $\boldsymbol{A}$ there is a boolean topos $\boldsymbol{B}$ and a faithful bicartesian representation $\boldsymbol{A} \to \boldsymbol{B}$* [2.542].

1.98 A **NATURAL NUMBERS OBJECT** (NNO) is $1 \xrightarrow{0} N \xrightarrow{s} N$ such that every $1 \xrightarrow{a} A \xrightarrow{t} A$ admits a unique $N \to A$:

$$\begin{tikzcd} 1 \arrow[r, "0"] \arrow[dr, "a"'] & N \arrow[r, "s"] \arrow[d, dashed] & N \arrow[d, dashed] \\ & A \arrow[r, "t"] & A \end{tikzcd}$$

Unique up to isomorphism. In $\mathcal{S}$: zero and successor. The topos of finite sets has no NNO.

1.981 *For a NNO and any $A \xrightarrow{x} B \xrightarrow{t} B$ there is a unique $A \times N \to B$:*

$$\begin{tikzcd} A \arrow[r, "\langle 1_A, 0\rangle"] \arrow[dr, "x"'] & A \times N \arrow[r, "1_A \times s"] \arrow[d, dashed] & A \times N \arrow[d, dashed] \\ & B \arrow[r, "t"] & B \end{tikzcd}$$

— transpose x, t into $1 \to B^A \to B^A$.

1.983 *Recursion: for a NNO and any $g : A \to B$, $h : A \times N \times B \to B$ there is a unique $f : A \times N \to B$ with $\langle 1_A, 0\rangle f = g$ and $(1_A \times s)f = \langle p_1, p_2, f\rangle h$.*

1.984 Addition, multiplication and exponentiation on a NNO by the usual recursion equations: $\langle 1_N, 0\rangle a = 1_N$, $(1_N \times s)a = as$; $\langle 1_N, 0\rangle m = 0$, $(1_N \times s)m = \langle p_1, m\rangle a$; $\langle 1_N, 0\rangle e = 0s$, $(1_N \times s)e = \langle p_1, e\rangle m$.

1.985 *For a NNO, $1 \xrightarrow{0} N \xleftarrow{s} N$ is a coproduct and $N \rightrightarrows N \to 1$ (by 1_N and s) is a coequalizer.*

$$\begin{tikzcd} 1 \arrow[dr, "0"'] & & N \arrow[dl, "s"] \\ & N & \end{tikzcd}$$

$$\begin{tikzcd} N \arrow[r, bend left, "1_N"] \arrow[r, bend right, "s"'] & N \arrow[r] & 1 \end{tikzcd}$$

1.986 *Given $1 \xrightarrow{a} A$ and $A \xrightarrow{f} A$ there is $A' \subset A$ carrying a', f' with*

$$\begin{tikzcd} 1 \arrow[r, "a'"] \arrow[dr, "a"'] & A' \arrow[r, "f'"] \arrow[d, tail] & A' \arrow[d, tail] \\ & A \arrow[r, "f"] & A \end{tikzcd}$$

and with $\binom{a'}{f'} : 1 + A' \to A$ *a cover* — in $\mathcal{R}e\ell(\boldsymbol{A})$ take $A'' = af^*$, $f'' = A''f$.

1.987 $1 \xrightarrow{a} A$, $A \xrightarrow{t} A$ has the **PEANO PROPERTY** iff every subobject $B \subset A$ that allows a and allows $t \restriction B$, the composite $B \subset A \xrightarrow{t} A$, is entire. *The A' of 1.986 is the least subobject allowing both a and $A' \subset A \xrightarrow{f} A$, and it has the Peano property.*

1.988 *In a boolean topos, if $1 \xrightarrow{a} A \xleftarrow{t} A$ is a coproduct and $A \rightrightarrows A \to 1$ (by 1_A and t) a coequalizer, then $1 \xrightarrow{a} A \xrightarrow{t} A$ has the Peano property.* ("Boolean" is removable by [2.542].)

1.989 *Under the hypotheses of 1.988, if $\binom{c}{u} : 1 + C \to C$ is a cover and $cf = a$, $uf = ft$, then $f : C \to A$ is monic.*

1.98(10) *Converse of 1.985: in any topos, if $1 \xrightarrow{a} A \xleftarrow{t} A$ is a coproduct and $A \rightrightarrows A \to 1$ a coequalizer, then $1 \xrightarrow{a} A \xrightarrow{t} A$ is a NNO.* Uniqueness from the Peano property applied to the equalizer of two candidates; existence from 1.986 applied to $\langle a, b\rangle$ and $t \times v$.

1.98(11) *A bicartesian functor $T : \boldsymbol{A} \to \boldsymbol{A}'$ of topoi carries a NNO to a NNO* — by the bicartesian characterization [1.985, 1.98(10)].

1.98(12) An **A-ACTION** is an object B with $1 \xrightarrow{b} B$ and $A \times B \xrightarrow{v} B$. A **FREE A-ACTION** is an A-action $A \times 1 \xrightarrow{1_A \times e} A \times A^* \xrightarrow{s} A^*$ universal among them:

$$\begin{tikzcd} A \times 1 \arrow[r, "1_A \times e"] \arrow[dr, "1_A \times b"'] & A \times A^* \arrow[r, "s"] \arrow[d, dashed, "1_A \times f"'] & A^* \arrow[d, dashed, "f"'] \\ & A \times B \arrow[r, "v"] & B \end{tikzcd}$$

Unique up to isomorphism; a NNO is a free 1-action. In $\mathcal{S}$, A^* is the set of finite lists, e the empty list, s prefixing.

1.98(13) *$A \times 1 \to A \times A^* \to A^*$ is a free A-action iff $\binom{e}{s}$: $1 + A \times A^* \to A^*$ is iso and $A \times A^* \rightrightarrows A^* \to 1$ is a coequalizer.*

1.98(14) *In a topos with a NNO every object A has a free A-action* — take A^* the least subobject of $(1+A)^N$ allowing e and $A \times A^* \subset A \times (1+A)^N \to (1+A)^N$.